Cryptography and Cybersecurity

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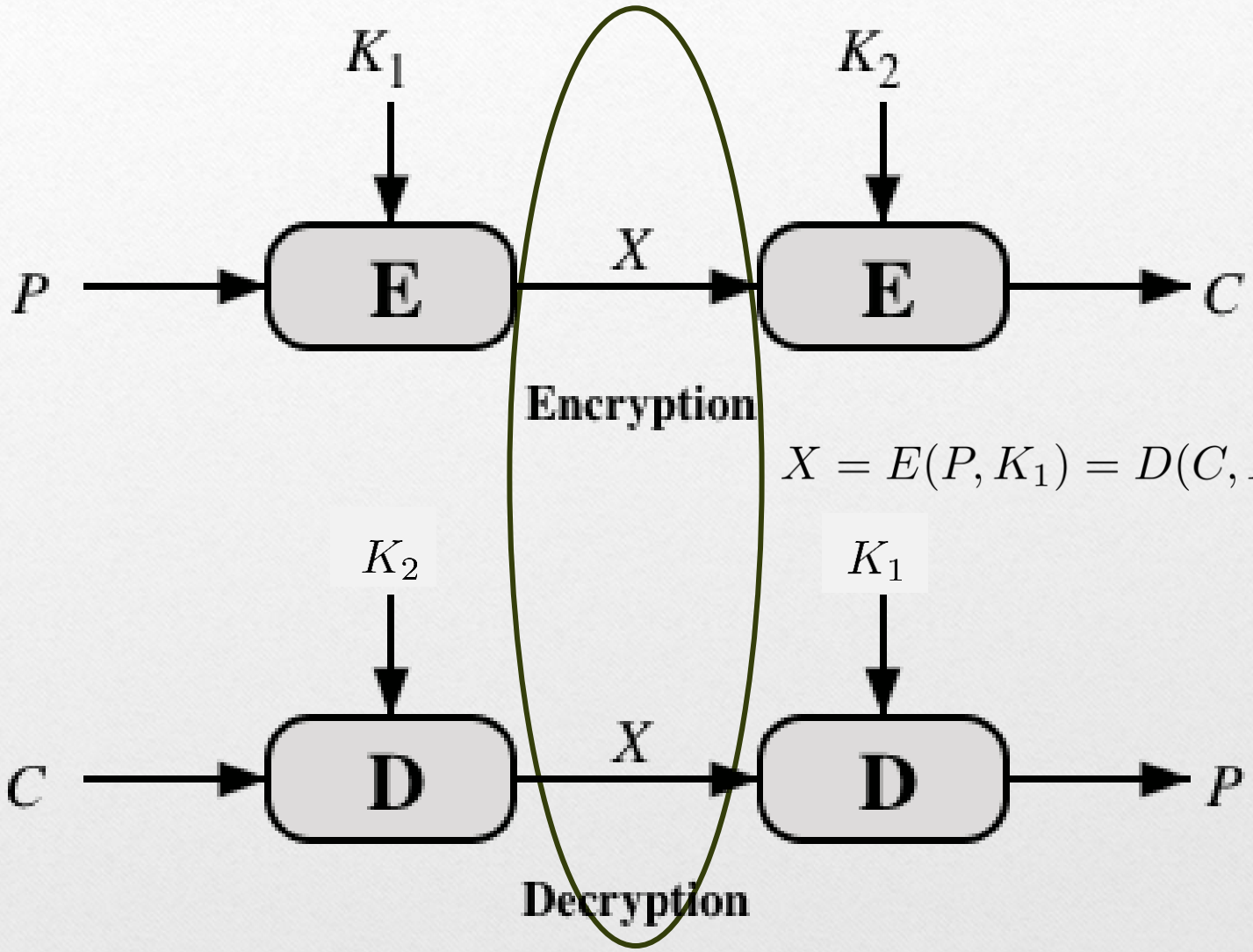
Homework 2:

Meet-in-the-Middle Attack



$$X = E(P, K_1) = D(C, K_2)$$

Meet-in-the-Middle Attack

Assume the pairs $(P_1, C_1), \ldots, (P_N, C_N)$ are available.

1. Consider the first pair (P_1, C_1). For all possible keys $K_{1,1}, \ldots, K_{1,M}$ and $K_{2,1}, \ldots, K_{2,M}$, compute

$$X_{1,1}^{(E)} = E(K_{1,1}, P_1), \ldots, X_{1,M}^{(E)} = E(K_{1,M}, P_1),$$

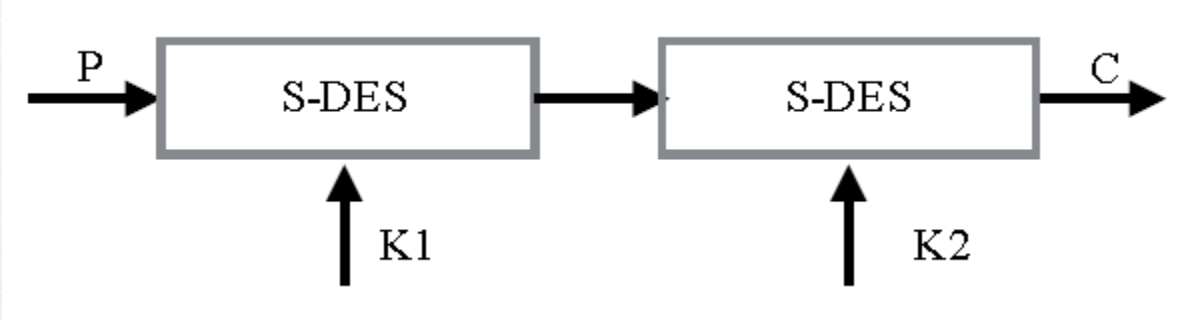
$$X_{2,1}^{(D)} = D(K_{2,1}, C_1), \ldots, X_{2,M}^{(D)} = D(K_{2,M}, C_1).$$

2. Look for all pairs (i, j) such that $X_{1,i}^{(E)} = X_{2,j}^{D}$. Each pair identifies a candidate pair $(K_{1,i}, K_{2,j})$ of possible keys.

3. Repeat Steps 1 and 2 using (P_2, C_2), but only for the key pairs identified in the previous step. This further reduces the number of candidate key pairs.

4. Iterate Steps 1 and 2 using each time a new pair (P_n, C_n), until only one pair of keys remains.

Meet-in-the-Middle Attack



Consider the above encryption scheme based on the simplified DES (S-DES) algorithm and using the MATLAB environment, write a code to implement the following points:

1. Assume $K1 = 123$ and $K2 = 1000$ and consider the plaintexts: $P1 = 182$, $P2 = 124$, $P3 = 87$. Generate the corresponding ciphertexts $C1$, $C2$, $C3$.

2. Based on the pairs (P1-C1), (P2-C2), (P3-C3), implement the meet-in-middle attack against the above scheme.